

Exp-1 Execute a Python Program in Google Cloud Shell

```
Welcome to Cloud Shell! Type "help" to get started.
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
You can view your projects by running `gcloud projects list`.
akulareddy072007@cloudshell:~$ nano hello.py
akulareddy072007@cloudshell:~$ python3 hello.py
Hello from Google Cloud!
Welcome to Cloud Computing
```

Exp-2 Create and Manage Files in Cloud Storage of Cloud Shell

```
Welcome to Cloud Computing
akulareddy072007@cloudshell:~$ mkdir cloudlab
mkdir: cannot create directory 'cloudlab': File exists
akulareddy072007@cloudshell:~$ cd cloudlab
akulareddy072007@cloudshell:~/cloudlab$ echo "Welcome to Google Cloud" > sample.txt
akulareddy072007@cloudshell:~/cloudlab$ cat sample.txt
Welcome to Google Cloud
akulareddy072007@cloudshell:~/cloudlab$ ls
hello.py  index.html  sample.txt
akulareddy072007@cloudshell:~/cloudlab$ cp sample.txt backup.txt
akulareddy072007@cloudshell:~/cloudlab$ ls
backup.txt  hello.py  index.html  sample.txt
akulareddy072007@cloudshell:~/cloudlab$
```

Exp-3. Run a Simple Web Server in Google Cloud Shell

Exp-4. System Resource Monitoring in Google Cloud Shell



Google Cloud Practical

My first cloud web application

```
top - 03:53:25 up 2:07, 0 user, load average: 0.00, 0.05, 0.28
Tasks: 45 total, 1 running, 44 sleeping, 0 stopped, 0 zombie
%Cpu(s): 1.0 us, 0.8 sy, 0.0 ni, 98.0 id, 0.0 wa, 0.0 hi, 0.2 si, 0.1 st
MiB Mem : 15998.9 total, 11892.6 free, 2446.6 used, 1938.5 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 13552.3 avail Mem
```

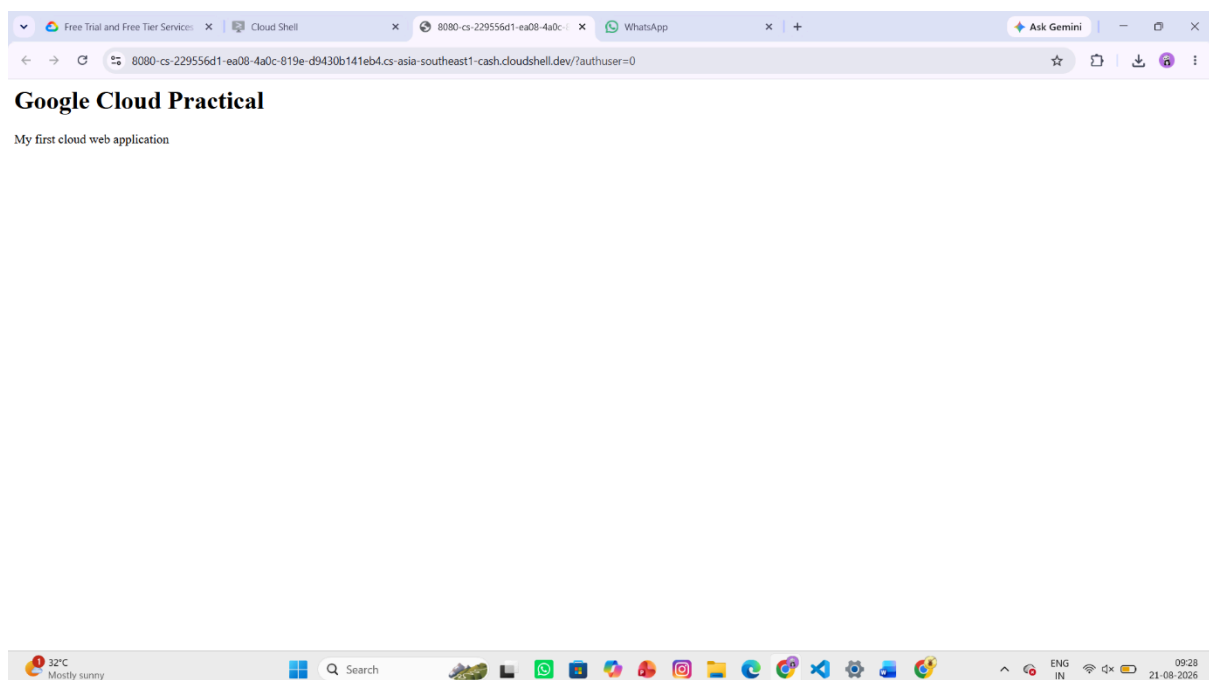
PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1638	akulare+	20	0	1475372	83728	48592	S	0.3	0.5	0:04.59	MainThread
2975	akulare+	20	0	18.5g	1.0g	63892	S	0.3	6.4	0:41.54	MainThread
4070	akulare+	10	-10	11852	5864	3684	R	0.3	0.0	0:00.01	top
1	root	20	0	4324	3288	3004	S	0.0	0.0	0:00.07	bash
9	syslog	20	0	222140	3836	2480	S	0.0	0.0	0:00.06	rsyslogd
25	root	20	0	38796	28148	10912	S	0.0	0.2	0:01.05	python
26	root	20	0	6196	2712	2472	S	0.0	0.0	0:00.00	logger
75	root	10	-10	12020	4048	2936	S	0.0	0.0	0:00.05	sshd
79	root	10	-10	12016	2864	1760	S	0.0	0.0	0:00.00	sshd
374	root	20	0	2099112	88084	64116	S	0.0	0.5	0:08.71	dockerd
451	root	20	0	1269404	13804	8516	S	0.0	0.1	0:01.15	editor-proxy
452	root	20	0	16328	6332	5408	S	0.0	0.0	0:00.01	sudo
465	root	20	0	1228132	3064	2200	S	0.0	0.0	0:00.00	tmux-agent
505	root	20	0	1894952	42512	29756	S	0.0	0.3	0:04.85	containerd
696	root	20	0	2696	1580	1484	S	0.0	0.0	0:00.00	sleep
750	akulare+	10	-10	7544	3088	2028	S	0.0	0.0	0:00.24	tmux
751	akulare+	10	-10	9312	6296	3928	S	0.0	0.0	0:00.04	bash
1146	root	20	0	9196	4160	3780	S	0.0	0.0	0:00.00	runuser
1147	akulare+	20	0	4324	3376	3108	S	0.0	0.0	0:00.00	sh
1169	akulare+	20	0	1781032	148336	52568	S	0.0	0.9	0:11.66	MainThread
1650	akulare+	20	0	8920	6100	4116	S	0.0	0.0	0:00.03	bash
2024	akulare+	10	-10	9312	6476	4108	S	0.0	0.0	0:00.04	bash
2476	akulare+	10	-10	9312	6424	4028	S	0.0	0.0	0:00.11	bash
2951	akulare+	10	-10	104004	20696	10780	S	0.0	0.1	0:00.41	python3
2960	root	10	-10	13992	9784	8404	S	0.0	0.1	0:00.01	sshd
2962	root	10	-10	13992	9716	8344	S	0.0	0.1	0:00.01	sshd
2964	akulare+	10	-10	14252	6368	4712	S	0.0	0.0	0:00.00	sshd
2965	akulare+	10	-10	14252	6444	4788	S	0.0	0.0	0:00.00	sshd
2966	akulare+	10	-10	7340	3640	3388	S	0.0	0.0	0:00.00	bash
2969	akulare+	10	-10	7604	4492	3896	S	0.0	0.0	0:00.00	bash
2972	akulare+	10	-10	7340	3596	3348	S	0.0	0.0	0:00.00	bash
2973	akulare+	10	-10	7340	3688	3428	S	0.0	0.0	0:00.00	start-shell.sh
2974	akulare+	10	-10	6876	3472	3172	S	0.0	0.0	0:00.00	tmux
2987	akulare+	20	0	1448728	63964	41904	S	0.0	0.4	0:00.33	MainThread

Exp-5. Execute a C Program in Google Cloud Shell

```
akulareddy072007@cloudshell:~$ nano add.c
akulareddy072007@cloudshell:~$ ./add
-bash: ./add: No such file or directory
akulareddy072007@cloudshell:~$ gcc add.c -o add
akulareddy072007@cloudshell:~$ nano add.c
akulareddy072007@cloudshell:~$ ./add
Sum = 30
akulareddy072007@cloudshell:~$
```

Exp-6. Simple Python Web Application – PaaS

```
Welcome to Cloud Shell! Type "help" to get started.
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
You can view your projects by running `gcloud projects list`.
akulareddy072007@cloudshell:~$ nano app.py
akulareddy072007@cloudshell:~$ python3 app.py
Traceback (most recent call last):
  File "/home/akulareddy072007/app.py", line 7, in <module>
    HTTPServer("", 8080).MyHandler().serve_forever()
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/socketserver.py", line 457, in __init__
    self.server_bind()
  File "/usr/lib/python3.12/http/server.py", line 136, in server_bind
    socketserver.TCPServer.server_bind(self)
  File "/usr/lib/python3.12/socketserver.py", line 473, in server_bind
    self.socket.bind(self.server_address)
OSError: [Errno 98] Address already in use
akulareddy072007@cloudshell:~$
```

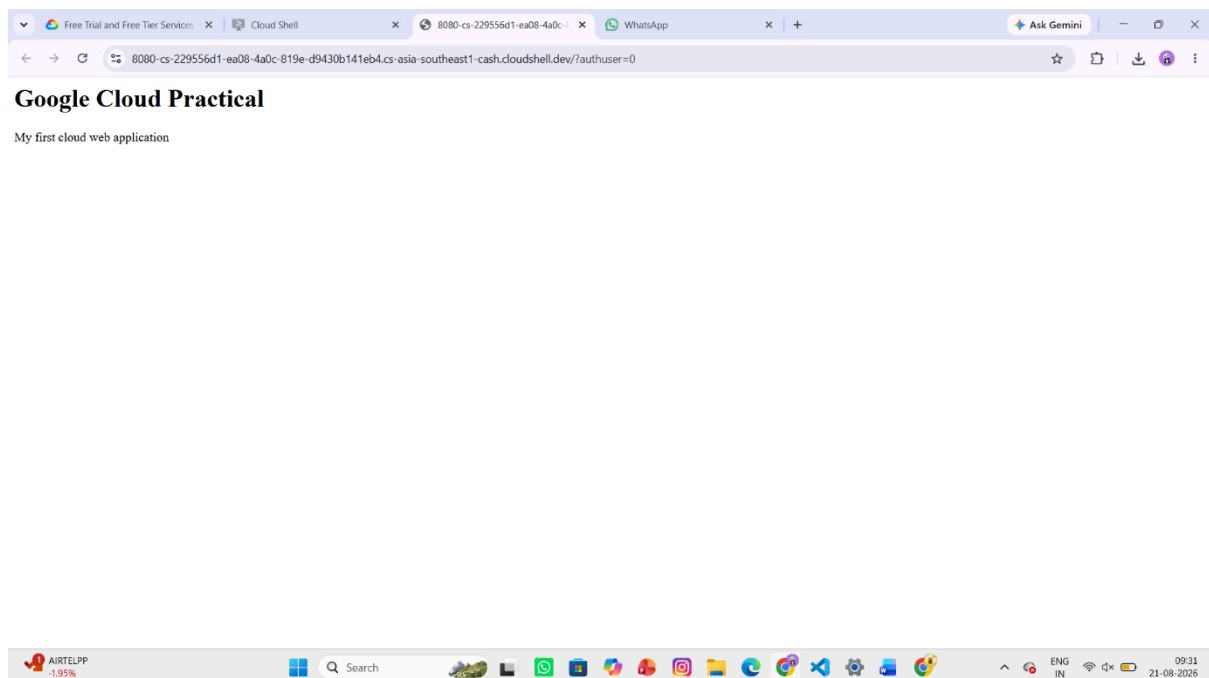


exp-7. Storage as a Service – Store and Access a File

```

OSError: [Errno 98] Address already in use
akulareddy072007@cloudshell:~/storage_demo$ mkdir storage_demo
akulareddy072007@cloudshell:~/storage_demo$ cd storage_demo
akulareddy072007@cloudshell:~/storage_demo/storage_demo$ echo "Welcome to Google Cloud Storage Demo" > student.txt
akulareddy072007@cloudshell:~/storage_demo/storage_demo$ cat student.txt
Welcome to Google Cloud Storage Demo
akulareddy072007@cloudshell:~/storage_demo/storage_demo$ python3 -m http.server 8080
Traceback (most recent call last):
  File "<frozen runpy>", line 198, in _run_module_as_main
  File "<frozen runpy>", line 88, in _run_code
  File "/usr/lib/python3.12/http/server.py", line 1314, in <module>
    test()
  File "/usr/lib/python3.12/http/server.py", line 1261, in test
    with ServerClass(addr, HandlerClass) as httpd:
    ~~~~~
  File "/usr/lib/python3.12/socketserver.py", line 457, in __init__
    self.server_bind()
  File "/usr/lib/python3.12/http/server.py", line 1308, in server_bind
    return super().server_bind()
    ~~~~~
  File "/usr/lib/python3.12/http/server.py", line 136, in server_bind
    socketserver.TCPServer.server_bind(self)
  File "/usr/lib/python3.12/socketserver.py", line 473, in server_bind
    self.socket.bind(self.server_address)
OSError: [Errno 98] Address already in use
akulareddy072007@cloudshell:~/storage_demo/storage_demo$ 

```



Exp-8. SQL Storage + Basic Query – DaaS

```
OSError: [Errno 98] Address already in use
akulareddy072007@cloudshell:~/storage_demo/storage_demo$ sqlite3 student.db
SQLite version 3.45.1 2024-01-30 16:01:20
Enter ".help" for usage hints.
sqlite> CREATE TABLE student(
  id INTEGER,
  name TEXT
);
sqlite>
sqlite> INSERT INTO student VALUES(1,'Anu');
INSERT INTO student VALUES(2,'Priya');
sqlite> SELECT * FROM student;
1|Anu
2|Priya
sqlite> 
```